🤌 XAI Lecture 02

Probing Further into ‘Interpretability’: Caveats & Challenges

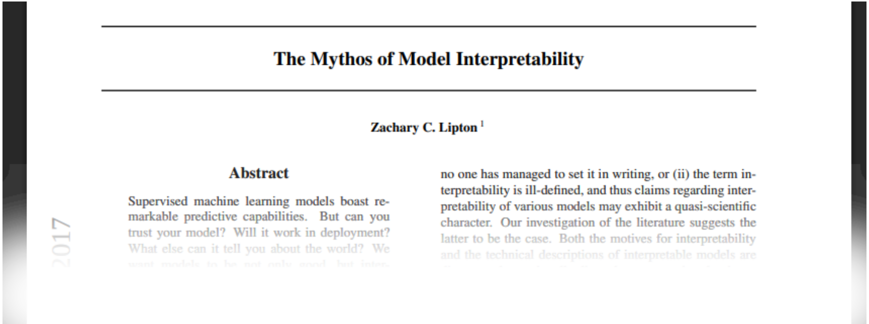
FAMAF - UNC

First Semester 2026

Agenda ---

- Two Papers:
  - Transparency: Motivations and Challenges
  - The Mythos of Model Interpretability
- Discussion

🧉 The Mythos of Model Interpretability ---



The Mythos of Model Interpretability

Zachary C. Lipton [1]

2017

Abstract

Supervised machine learning models boast remarkable predictive capabilities. But can you trust your model? Will it work in deployment? What else can it tell you about the world? We

no one has managed to set it in writing, or (ii) the term interpretability is ill-defined, and thus claims regarding interpretability of various models may exhibit a quasi-scientific character. Our investigation of the literature suggests the latter to be the case. Both the motives for interpretability and the technical descriptions of interpretable models are

(Lipton 2018)

🧉 Contributions ---

- **Goal:** Refine the discourse on interpretability
- Outline **desiderata of interpretability research**
  - Motivations for interpretability are often diverse and discordant
- Identifying **model properties and techniques** thought to confer interpretability

🧉 Motivation ---

- We want models to be **not only good** w.r.t. predictive capabilities, **but also interpretable**
- Interpretation is **underspecified**
  - Lack of a formal technical meaning
- Papers provide **diverse and non-overlapping motivations** for interpretability

🧉 Prior Work: Motivations for Interpretability ---

Interpretability promotes trust

- But **what is trust?**
- Is it faith in model performance?
- If so, why are accuracy and other standard performance evaluation techniques inadequate?

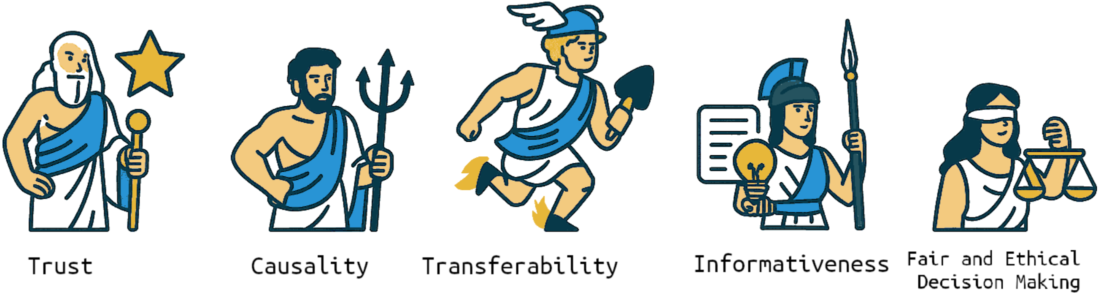
🧉 When is interpretability needed? ---

- Simplified optimization **objectives fail to capture complex real life goals**
  - Algorithm for hiring decisions – productivity **and** ethics
  - Ethics is hard to formulate
- Training data is **not representative of deployment environment**

Interpretability serves those objectives that we deem important but struggle to model formally!

🧉 Desiderata ---

Understanding motivations for interpretability through the lens of prior literature:



🧉 Desiderata: Trust ---

- Is trust simply confidence that the model will perform well?
- A person might **feel at ease with a well understood model**, even if this understanding has no purpose
- **Training and deployment objectives diverge**
  - E.g., model makes accurate predictions but not validated for racial biases
- **Trust $\rightarrow$ relinquish control**
  - For which examples is the model right?

Desiderata: Causality ---

- Researchers hope to **infer properties** (beyond correlational associations) from **interpretations/explanations**
  - Regression reveals strong association between smoking and lung cancer
- However, **task of inferring causal relationships from observational data is a field in itself**
  - Judea Pearl (*left*)
  - Don Rubin (*right*)

🧉 Desiderata: Transferability ---

- Humans exhibit **richer capacity to generalize, transferring learned skills** to unfamiliar situations
  - Model's generalization error: gap between performance on training and test data
  - We already use ML in non-stationary environments

- **Environment might even be adversarial**
  - Changing pixels in an image tactically could throw off models but not humans

- **Predictive models can often be gamed**
  - In such cases, predictive power loses meaning

Desiderata: Informativeness ---

- **Predictions → Decisions**
  - Convey additional information to human decision makers

- **Example:** Which conference should I target?
  - A one word answer is not very meaningful

- **Interpretation might be meaningful even if it does not shed light on model's inner workings**
  - Similar cases for a doctor in support of a diagnosis

🧉 Desiderata: Fair & Ethical Decision Making ---

- **ML is being deployed in critical settings**
  - E.g., healthcare

- How can we be sure **algorithms do not discriminate** on the basis of race?
  - AUC is not good enough

Side note

European Union enforces the **Right to explanation**

🧉 Properties of Interpretable Models ---

- **Transparency**
  - How exactly does the model work?
  - Details about its inner workings, parameters etc.

- **Post-hoc explanations**
  - What else can the model tell me?
  - E.g., visualizations of learned model, explaining by example

Transparency: Simulatability ---

- **Can a person contemplate the entire model at once?**
  - Need a very simple model

- A human should be able to **take input data and model parameters and calculate prediction**

🧉 Transparency: Decomposability ---

- **Understanding each input, parameter, calculation**
  - E.g., decision trees, linear regression

- **Inputs must be interpretable**
  - Models with highly engineered or anonymous features are not decomposable

🧉 Transparency: Algorithmic Transparency ---

- **Learning algorithm itself is transparent**
  - E.g., linear models (error surface, unique solution)

- Modern **deep learning methods lack this kind of transparency**
  - We don't understand how the optimization methods work
  - No guarantees of working on new problems

Note:

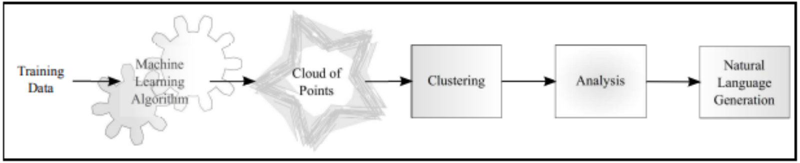
Humans do not exhibit any of these forms of transparency

🧉 Post-hoc: Text Explanations ---

- Humans often justify decisions verbally (post-hoc)

In **Learning from explanations using sentiment and advice in RL** (Krening et al. 2016)

- **Krening et. al. 2016:** {Learning from explanations using sentiment and advice in RL}
  - One model is a reinforcement learner
  - Another model maps models states onto verbal explanations
  - Explanations are trained to maximize likelihood of ground truth explanations from human players
  - So: **explanations do not faithfully describe agent decisions, but rather human intuition**



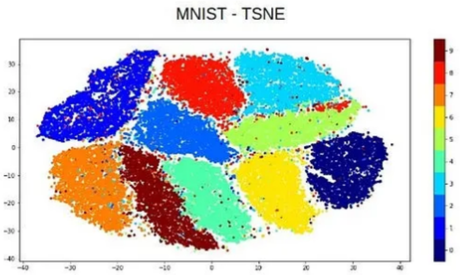
Thoughtland's architecture from

(Duboue 2013)

🧉 Post-hoc: Visualization ---

- **Visualize high-dimensional data with t-SNE**
  - 2D visualizations in which nearby data points appear close

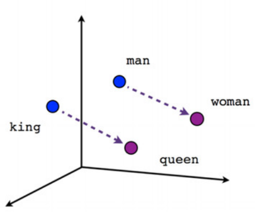
- Perturb input data to enhance **activations of certain nodes in neural nets** (image classification)
  - Helps understand which nodes corresponds to what aspects of the image
  - E.g., certain nodes might correspond to dog faces



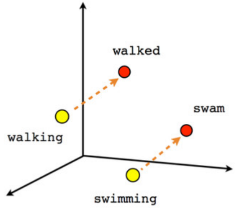
Post-hoc: Example Explanations ---

- Reasoning with **examples**
  - E.g., Patient A has a tumor because he is similar to these k other data points with tumors

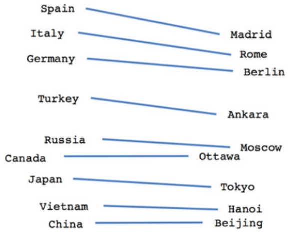
- k-neighbors can be computed by using some distance metric on learned representations
  - E.g., word2vec (Mikolov et al. 2013)



Male-Female



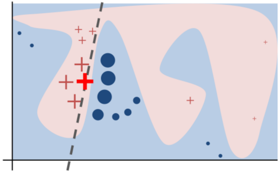
Verb tense



Country-Capital

🧉 Post-hoc: Local Explanations ---

- **Hard to explain a complex model in its entirety**
  - How about explaining **smaller regions**? (Ribeiro, Singh, and Guestrin 2016)



- Explains decisions of any model in a local region around a particular point
- Learns sparse linear model

🧉 Transparency and Post-hoc Key Insights ---

Claims about interpretability must be qualified

- If a model satisfies a form of transparency, highlight that clearly
- For post-hoc interpretability, fix a clear objective and demonstrate evidence

Transparency may be at odds with broader objectives of AI

Example: Health care

- Choosing interpretable models over accurate ones **to convince decision makers**
- Short term goal of building trust with doctors **might clash** with long term goal of improving health care

🧉 Post-hoc interpretations can mislead ---

Do not blindly embrace post-hoc explanations!

- Post-hoc explanations can seem **plausible but be misleading**
  - They do not claim to open up the black-box
  - They only provide plausible explanations for its behavior
  - E.g., text explanations

🧉 Summary: The Mythos of Model Interpretability ---

- **Goal:** Refine the discourse on interpretability

- Outline **desiderata of interpretability research**
  - Motivations for interpretability are often diverse and discordant

- Identifying **model properties and techniques** thought to confer interpretability

Transparency: Challenges and Motivation ---

Transparency: Motivations and Challenges*

Adrian Weller[1,2,3][0000–0003–1915–7158]

[1] University of Cambridge, UK, adrian.weller@eng.cam.ac.uk
[2] The Alan Turing Institute, UK
[3] Leverhulme Centre for the Future of Intelligence, UK

19 Aug 2019

Abstract. Transparency is often deemed critical to enable effective real-world deployment of intelligent systems. Yet the motivations for and benefits of different types of transparency can vary significantly depending on context, and objective measurement criteria are difficult to identify. We provide a brief survey, suggesting challenges and related concerns, particularly when agents have misaligned interests.
We highlight and review settings where transparency may cause harm, discussing connections across privacy, multi-agent game theory, eco-

(Weller 2019)

🧉 Contributions ---

- Characterizing different kinds of transparency, and underlying motivations
- Shedding light on the downsides of having transparency

🧉 Types and Goals of Transparency ---

- **Type 1:** For a developer, to understand how their system is working, aiming to debug or improve it: to see what is working well or badly, and get a sense for why (`Internal Debugging`).
- **Type 2:** For a user, to provide a sense for what the system is doing and why, to enable prediction of what it might do in unforeseen circumstances and build a sense of trust in the technology (`User Trust`).
- **Type 3:** For society broadly to understand and become comfortable with the strengths and limitations of the system, overcoming a reasonable fear of the unknown (`Social Acceptance`).

I made up the names myself 😎

🧉 Types and Goals of Transparency (cont.) ---

- **Type 4:** For a user to understand why one particular prediction or decision was reached, to allow a check that the system worked appropriately and to enable meaningful challenge (e.g. credit approval or criminal sentencing) (**Individual Explanation**).
- **Type 5:** To provide an expert (perhaps a regulator) the ability to audit a pre-diction or decision trail in detail, particularly if something goes wrong (e.g. a crash by an autonomous car). This may require storing key data streams and tracing through each logical step, and will facilitate assignment of accountability and legal liability (**Expert Audit**).

I made up the names myself 😎

🧉 Types and Goals of Transparency (cont.) ---

- **Type 6:** To facilitate monitoring and testing for safety standards (`Safety Monitoring`).
- **Type 7:** To make a user (the audience) feel comfortable with a prediction or decision so that they keep using the system. Beneficiary: deployer (`User Comfort`).
- **Type 8:** To lead a user (the audience) into some action or behavior (`Behavioral Influence`).
  - e.g. Amazon might recommend a product, providing an explanation in order that you will then click through to make a purchase. Beneficiary: deployer.

I made up the names myself 😎

Transparency: Global vs. Local ---

- **Global**
  - Understanding whole system
  - Types 2 (User Trust) – 3 (Social Acceptance)
- **Local**
  - Explanation for a particular prediction
  - Types 4 (Individual Explanation), 5 (Expert Audit), 6 (Safety Monitoring), 7 (User Comfort), 8 (Behavioral Influence)

🧉 Key Challenges ---

Explanations are beneficial to the society only if they are faithful

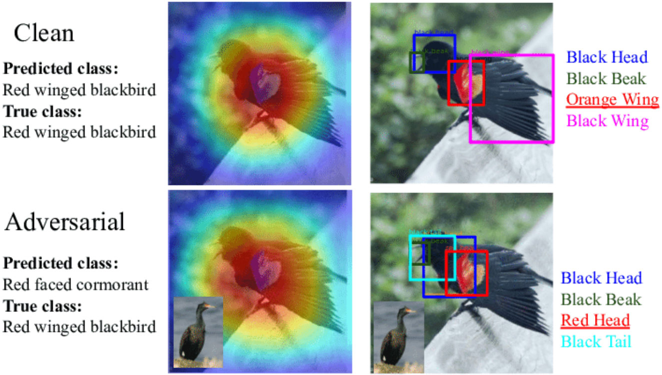
- Defining criteria and tests for practical faithfulness are important open problems
  - **Context is important!**

Understanding and Communication is hard

- Is an explanation good at conveying faithful information in understandable form, and if a human has actually understood it well?
- Context dependent
- Need for deeper probing!

🧉 Key Challenges (cont.) ---

Comparing Explanations is HARD!



If we have different saliency maps, how to know which one is better

(Gulshad and Smeulders 2020)

🧉 Possible Dangers: Audience vs. Beneficiary ---

- **Recommender systems**
  - Amazon (Beneficiary) and its Users (Audience)
- **Healthcare**
  - Google Verily (Beneficiary) and Hospitals/Doctors (Audience)
- **Criminal Justice**
  - COMPAS (Beneficiary) and Courts/Judges (Audience)

🧉 Posible Dangers: Government Use of Algorithms (cont.) ---

- COMPAS system predicts risk of recidivism
- A prisoner should have some transparency into the decision made by COMPAS
  - To ensure **proper process has been followed**
  - Enable potential challenge
- But, **can there be too much transparency?** (Next topic)
- Also, recent push for making all code/data for such models public. Good idea?

COMPAS Violent recidivism risk scale (Wikipedia version)

The violent recidivism score is meant to predict violent offenses following release. The scale uses data or indicators that include a person's "history of violence, history of non-compliance, vocational/educational problems, the person's age-at-intake and the person's age-at-first-arrest (Wikipedia contributors 2025)

🧉 Posible Dangers: Government Use of Algorithms (cont.) ---

The violent recidivism risk scale is calculated as follows:

$$s = a(-w) + a_{\text{first}}(-w) + h_{\text{violence}}w + v_{\text{edu}}w + h_{\text{nc}}w$$

where:

s is the violent recidivism risk score, w is a weight multiplier , a is current age, $a_{\textbf{first}}$ is the age at first arrest, $h_{\textbf{violence}}$ is the history of violence, $v_{\textbf{edu}}$ is vocational education scale, and $h_{\textbf{nc}}$ is history of noncompliance

The weight, w, is *"determined by the strength of the item's relationship to person offense recidivism that we observed in our study data."* (Wikipedia contributors 2025)

🧉 Posible Dangers: Gaming, IP Incentives, and Privacy ---

- If all details available, the process can be **gamed**
- Less incentive for private IP and slow progress
- Privacy and transparency are often in **conflict**
  - How much transparency is too much in a setting?

🧉 Means and Ends ---

- Transparency → reliability, fairness
- If we are able to develop a good set of "safety checks", then may be it is ok to not have full transparency
  - E.g., autonomous vehicles

"Does your car have any idea why my car pulled it over?"

🧉 Societal Considerations ---

Does giving more information to each individual help society?

Braess' paradox

More information empowers the agents to optimize their own agendas more efficiently, and thus may lead to a worse global outcome

🧉 Selective Transparency & Discrimination ---

- Selective Transparency may hurt people disproportionately
- Furthermore, transparency about certain attributes (e.g., gender) in certain settings is known to cause discriminatory behavior as well

🧉 References --- I

Duboue, Pablo. 2013. "On the Feasibility of Automatically Describing N-Dimensional Objects." In *Proceedings of the 14th European Workshop on Natural Language Generation*, 172–77.

Gulshad, Sadaf, and Arnold Smeulders. 2020. "Explaining with Counter Visual Attributes and Examples." In *Proceedings of the 2020 International Conference on Multimedia Retrieval*, 35–43.

Krening, Samantha, Brent Harrison, Karen M Feigh, Charles Lee Isbell, Mark Riedl, and Andrea Thomaz. 2016. "Learning from Explanations Using Sentiment and Advice in Rl." *IEEE Transactions on Cognitive and Developmental Systems* 9 (1): 44–55.

Lipton, Zachary C. 2018. "The Mythos of Model Interpretability: In Machine Learning, the Concept of Interpretability Is Both Important and Slippery." *Queue* 16 (3): 31–57.

Mikolov, Tomas, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. "Efficient Estimation of Word Representations in Vector Space." *arXiv Preprint arXiv:1301.3781*.

Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. 2016. "" Why Should I Trust You?" Explaining the Predictions of Any Classifier." In *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, 1135–44.

🧉 [--- ||

🧉 References ---]🧉 References ---

Weller, Adrian. 2019. "Transparency: Motivations and Challenges." In *Explainable Ai: Interpreting, Explaining and Visualizing Deep Learning*, 23–40. Springer.

Wikipedia contributors. 2025. "COMPAS (Software) — Wikipedia, the Free Encyclopedia." https://en.wikipedia.org/w/index.php?title=COMPAS_(software)&oldid=1303968797.